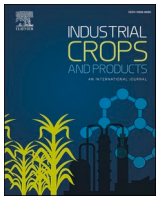
Industrial Crops & Products 216 (2024) 118730



Contents lists available at ScienceDirect

Industrial Crops & Products

journal homepage: www.elsevier.com/locate/indcrop





A deep eutectic solvent pretreatment with self-cleaning lignin droplets function to efficiently improve the enzymatic saccharification and ethanol production of bamboo residues

Qinpei Hou [c], Zhaoming Liu [c], Zhengjun Shi [a,b], Haiyan Yang [a,b,c], Dawei Wang [b,c], Jing Yang [a,b,c,*]

[a] Key Laboratory for Forest Resources Conservation and Utilization in the Southwest Mountains of China, Ministry of Education, Southwest Forestry University, Kunming 650224, China
[b] Key Laboratory for Yunnan Province Universities on Chemical Utilization of Forestry Lignocellulosic Biomass, Education of Department of Yunnan Province, Kunming 650224, China
[c] School of Materials and Chemical Engineering, Southwest Forestry University, Kunming 650224, China

ARTICLE INFO

Keywords:
Bamboo residues
Deep eutectic solvent pretreatment
Lignin droplets
Enzymatic hydrolysis
Ethanol production

ABSTRACT

Bamboo is considered a promising biomass resource for solving the energy crisis and climate change. Methacryloxyethyltrimethyl ammonium chloride (DMC) as a hydrogen bond acceptor, was used to synthesize a new deep eutectic solvent (DES) with self-cleaning lignin droplets function during the bamboo pretreatment. When *D. giganteus* were pretreated with DES of DMC and oxalic acid (OA) at 130 °C for 4 h, about 96% of xylan and 81% of lignin were removed from the bamboo wastes, and 54.44% of glucose yield and 13.5 g/L of ethanol production were achieved. Compared with the native lignin from raw bamboo, the molecular weight and β-O-4 aryl ethers in the regenerated lignin fractions were significantly decreased. In addition, the DMC/OA pretreatment reduced the lignin droplets and surface lignin content of pretreated substrates by comparison with choline chloride (ChCl)based DES pretreatment, leading to more cellulose substrates exposed. The enzymatic digestibility of DMC/OA-treated bamboo was 4 times higher than that of ChCl/OA. This work provided insights to the discovery of more functional DESs for the biorefinery of lignocellulosic biomass.

1. Introduction

Bamboo, due to its fast growth rate (3–5 years) and chemical composition similar to wood, has become a promising biomass resource in biorefinery system (Scurlock, et al., 2000). *Dendrocalamus giganteus Munro* are planted widely in the southwest of China, and it was used as feedstock in the food, textile, construction, and papermaking industry. During these processes, approximately 3 million tons or more of bamboo processing residue are left annually (Liang et al., 2023). These bamboo wastes can be processed into sugar-based compounds and bioethanol through chemical or biochemical methods, which can reduce waste management and environmental pollution issues, and also provide sustainability for energy resources.

Pretreatment technology was required to destroy the native recalcitrant structure of bamboo, making it easier for the carbohydrate fraction to hydrolyze into monosaccharides, and then ferment into ethanol (Ahmed et al., 2022; Anu et al., 2020). Deep eutectic solvent (DES) prepared with hydrogen bond donors (HBD) and acceptors (HBA) at proper ratios has attracted much interests of researcher due to ease of synthesis, less toxic, biodegradable, and low cost (Ahmed et al., 2022; Lugani et al., 2020). The biomass pretreatment of DES can efficiently remove non-crystalline complexes, such as hemicellulose and lignin, by breaking the lignin-carbohydrate linkages and ether bonds of lignin matrix (Jose et al., 2023; Areeya et al., 2023). Choline chloride (ChCl) was widely adopted as HBA in the synthesis of DES for biomass pretreatment. A detailed discussion on ChCl-based DES pretreatment of lignocellulose have been proposed. Many studies have shown that the ChCl-based DES was a good solvent for lignin solubility and choline chloride increased the β-O-4 cleavage rate of milled wood lignin (Smink et al., 2019; Chen et al., 2020; Shen et al., 2020). The chloride ions can

* Correspondence to: Key Laboratory for Forest Resources Conservation and Utilization in the Southwest Mountains of China, Ministry of Education, Southwest Forestry University, 300 Bailong Lu, Kunming 650224, China,
E-mail address: kmjingyang@163.com (J. Yang).

https://doi.org/10.1016/j.indcrop.2024.118730
Received 8 February 2024; Received in revised form 23 April 2024; Accepted 9 May 2024
Available online 16 May 2024
0926-6690/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

help cleave lignin carbohydrate complexes (LCCs) of biomass, and the formation of new strong hydrogen bonds between Cl - in DES and hydroxyl groups in carbohydrates and lignin, can weaken the hydrogen bond network inside lignocellulose, which can result in the dissolution of biomass components (Liu et al., 2017; Wang et al., 2020). In addition, Li's group have reported that the hydroxyl groups of lignin could undergo nucleophilic substitution by chloride anions, resulting in the formation of stable intermediates that facilitate the cleavage of lignin ether bonds (Li et al., 2018).

Although the ChCl-based DES system has shown excellent performance in lignocellulose fractionation, other HBAs with special functions have also attracted the attention of some researchers, which had high biomass pretreatment performance and special functions in cracking the LCC linkages of biomass (Selkälä et al., 2016; Yang et al., 2022; Tang et al., 2023). Recently, Tang et al. (2023) reported a new type of DES with additional functions using CTAB and LA. They found that CTAB not only prevented the deposition and formation of pseudo lignin, but solubilized some waxes on the substrates, significantly improving the accessibility of pretreated rice husks. Our previous results showed that DES pretreatment of tetramethylammonium chloride and oxalic acid can produce some lignin droplets on the pretreated substrates, which reduced the degradability of bamboo. These lignin droplets that affect carbohydrate bioconversion can be removed through washing the pretreated residues with dilute alkaline solution (Yang et al., 2022). Methacryloxyethyltrimethyl ammonium chloride (DMC), as an active monomer with unsaturated ethylene and quaternary ammonium salt groups, is widely used in the synthesis of various functional polymers, for example, polymerization of methacryloxyethyltrimethyl ammonium chloride (PDMC). The synthesis process of DMC has the advantages of low cost, low energy consumption, and ease of preparation (Deng et al., 2011; Wang et al., 2016). Therefore, DMC, as a quaternary ammonium salt with longer alkyl chains, can also serve as a HBA in DES synthesis to form hydrogen bonds with hydrogen bond donors (HBD).

The main objective of this study was to develop an innovative DMC-based DES pretreatment with self-cleaning functions, which can prevent the deposition and formation of pseudo lignin and lignin droplets during the bamboo pretreatment. The process parameters of DMC/OA pretreatment were optimized and discussed in terms of the removal of lignin and xylan, as well as enzymatic digestibility and bioethanol production. And complete characterizations (BET surface area, XRD, SEM, XPS, hydrophobicity, and FTIR technology) were further performed on the substrates to explain the mechanism of DMC-based DES pretreatment. And the native lignin in bamboo and regenerated DES lignin fraction were collected; the GPC and 2D-HSQC NMR techniques were used to characterize their chemical structure. Finally, the hydrogen bond number and interaction energy were measured to further explore the interaction mechanism between DMC-based DES system and bamboo biomass.

2. Materials and methods

2.1. Materials

Dendrocalamus giganteus Munro (*D. giganteus*, DG) harvested from Yunnan Province, China, was milled into bamboo powder (40–60 mesh) and dried in an oven at 40 °C for 10 h. Cellulase enzyme (CellicCTec2 with 220 FPU/mL and Celluclast 1.5 L with protein content of 40 mg/mL) was supplied by Sigma-Aldrich (Shanghai). *Saccharomyces cerevisiae* from Yichang Angel Yeast Company was used during the ethanol fermentation.

2.2. DES pretreatment of bamboo residues

The DMC (HBA) and oxalic acid (HBD) of different molar ratio were mixed at 90 °C until a homogeneous liquid formed. The Kamlet-Taft empirical parameters (α, β, and π*) were adopted to characterize these DES systems (Xue et al., 2016). Bamboo powder was pretreated with DMC-based DES at 110–140 °C and various times (3–8 h) with stirring of 150 rpm. The solid/liquid ratio of DG and DES was 1:10 (g/mL). After the pretreatment, the mixture was filtered using Brinell funnel, and the solid fractions were washed until neutral. The bamboo pretreated with DES was dried overnight in the oven of 40 °C for enzymatic hydrolysis and ethanol fermentation. The liquid fraction was acidified to pH 2 with sulfuric acid to precipitate lignin. The lignin obtained was referred to as DMC/OA lignin. And the supernatant was collected for the DES recovery. The DES pretreated samples under the different conditions were named as DMC/OA-110 for the pretreatment temperature of 110 °C, DMC/OA-3 for pretreatment temperature of 3 h, and DMC/OA-1:1 for the DMC/OA ratio of 1:1, respectively.

At the same time, bamboo powder pretreated with DES of ChCl/OA or liquid water hot under (LWH) a ratio of 1:10 (w/v) and 130 °C for 4 h was taken as control. The treated bamboo powder was labeled as ChCl/OA-DG and LHW-DG, respectively. The NREL standard method was used to analyze the chemical composition of untreated and pretreated bamboo (Sluiter et al., 2011). The removal of xylan, delignification and glucan recovery after pretreatment were obtained by the formulas given below:

$$\text{Xylan removal}\ (\%) = 1 - \frac{\text{xylan in treated bamboo}\ (g)}{\text{xylan in the raw biomass}\ (g)} \times 100 \quad (1)$$

$$\text{Delignification}(\%) = 1 - \frac{\text{lignin in treated bamboo}(g)}{\text{lignin in the raw biomass}(g)} \times 100 \quad (2)$$

$$\text{Glucan recovery}(\%) = \frac{\text{glucan in treated bamboo}\ (g)}{\text{glucan in the raw biomass}\ (g)} \times 100 \quad (3)$$

2.3. Enzymatic saccharification and fermentation of bamboo wastes

Enzymatic hydrolysis of samples was operated in an air incubator (150 rpm and 50 °C) at 2 wt% solid concentration and 50 mM sodium citrate buffer solution for 72 h. The cellulase (Cellic® CTec2) was loaded at 20 FPU/g biomass. The hydrolysate was centrifuged and then analyzed the content of reducing sugars using high-performance liquid chromatography (HPLC). On the other hand, Tween 80 of 150 mg/g substrates was added to further enhance the yield of glucose and xylose during the enzymatic hydrolysis.

The enzymatic hydrolysate (16 mL) and *Saccharomyces cerevisiae* (3 g/L) were mixed at 35 ℃ for 24 h. The content of sugars and ethanol in the fermented broth were analyzed by HPLC (Agilent Technologies, 1260 series, USA), equipped with a Bio-Rad HPX-87 H column. The enzymatic hydrolysis efficiency and bioethanol production were calculated, in accordance with our previous publications (Meng et al., 2023).

2.4. Cellulase adsorption study of treated biomass

Enzymatic adsorption capacity was determined at 4 ℃ and 150 rpm by combining 10 mg of bamboo powder with the concentration range of (0.01–2) mg/mL cellulase solution (Yang et al., 2022; Meng et al., 2023). The adsorption isotherms with different enzymatic concentrations were studied by UV–vis spectrophotometry after 3 h adsorption balance. The enzymatic adsorption capacity on solid residue were obtained by Langmuir isothermal adsorption equation ($\Gamma = \Gamma_{max}\,KC/(1+KC)$), Γ and Γ_{max} (mg/g) are the equilibrium and maximum adsorption capacity, respectively. C (mg/mg) the concentration of enzyme protein in the liquid. K (mL/mg cellulase) the constant of Langmuir. And the R (R=Γ max ×K) is the distribution coefficient (L/g). All pretreatment, enzyme hydrolysis and adsorption experiments were performed in triplicate.

2.5. Structural characterization analysis of pretreated bamboo residues

The hydrophobicity and accessibility of DG were determined using Rose Bengal dye and Congo red dye (DR 28) adsorption method (Gessner et al., 2000; Inglesby and Zeronian, 2002). The average DPs were evaluated using copper ethylene-diamine solvent method (Lin et al., 2021). Using the Ultima IV XRD instrument (Rigaku Corporation, Japan) to analyze crystallinity index (CrI) of the substrates, and the CrI/cellulose content was used to evaluate the actual effect of DES pretreatment on CrI (Pihlajaniemi et al., 2016). FTIR was determined using the KBr press method with a sample to KBr mass ratio of 1:10 and analyzed by 710 FTIR spectrophotometer (Nicolet, USA) over 4000–400 cm^{-1}. Dry samples were sprayed with gold powder for 30–90 s, and then adhered to conductive adhesive for SEM analysis (Sigma 300, Germany,10 kv). The relative surface area was measured by Micromeritics ASAP2460 analyzer instrument (USA). XPS spectra were obtained on an Escalab 250Xi spectrometer (Thermo Fisher Scientific Inc., US), and the surface coverage by lignin (S_{lig}) and carbohydrates (S_{carb}) were estimated using the research of Meng et al. (2023). The weight of number-average molecular (Mn) and weight-average molecular (Mw) for willed wood lignin (MWL)and DES-lignin samples were obtained using Gel Permeation Chromatography (GPC) analysis (Waters E2695, America). Additionally, a Bruker AVIII 600 MHz M instrument was used to conduct the nuclear magnetic resonance (NMR) spectra of the lignin samples.

The optimization of molecular structures was achieved using the ORCA program, which employed the density functional B3LYP D3 method and the def2-TZVP basis set (Frank, 2022). In addition, the electrostatic potential distribution (ESP) of DES was analyzed using the Multiwfn program, and the VMD program was utilized for visualization purposes (Lu and Chen, 2012, Humphrey et al., 1996). Classic molecular dynamics simulations were carried out to investigate the mixed solution from the atomic level. Cellobiose, 4-O-methyl Gluconic acid xylose and Veratrylglycerol-β-guaiacyl ether (VG) was selected as the cellulose, hemicellulose and lignin model, respectively. Kinetic simulations of cellulose-hemicellulose-lignin without DES were first conducted as a control group, and then the two DES systems were added. The number of hydrogen bonding, interaction energy and free volume of these system during the kinetic simulations were analyzed using GROMACS 2020 program .

3. Results and discussion

3.1. Evaluation of DMC/OA pretreatment efficiency

The pretreatment performance was evaluated according to xylan and lignin removal, and release of fermentable sugars in Fig. 1. The raw DG contained 47.42% of glucan, 16.61% of xylan, 28.31% of lignin, which was carried out the DMC/OA pretreatment under different conditions. It was found that the solid residue recovery by the DMC/OA pretreatment decreased to about 60% in Fig. 1a-c, indicating that part of the raw bamboo has been dissolved. As seen in Figure1a, the molar ratio of DMC and OA especially affected the DES pretreatment performance. As the molar ratio of DMC and OA increased from 1:1 to 3:1, the removal of xylan was 96.88%, 96.99% and 94.52% respectively, without significant changes. It is noteworthy that an obvious increase in delignification (79.56%, 81.53% and 87.52%) were observed, suggesting that the dosage of DMC played a critical role in improving the solubility and degradability of lignin. When these pretreated substrates were tested for

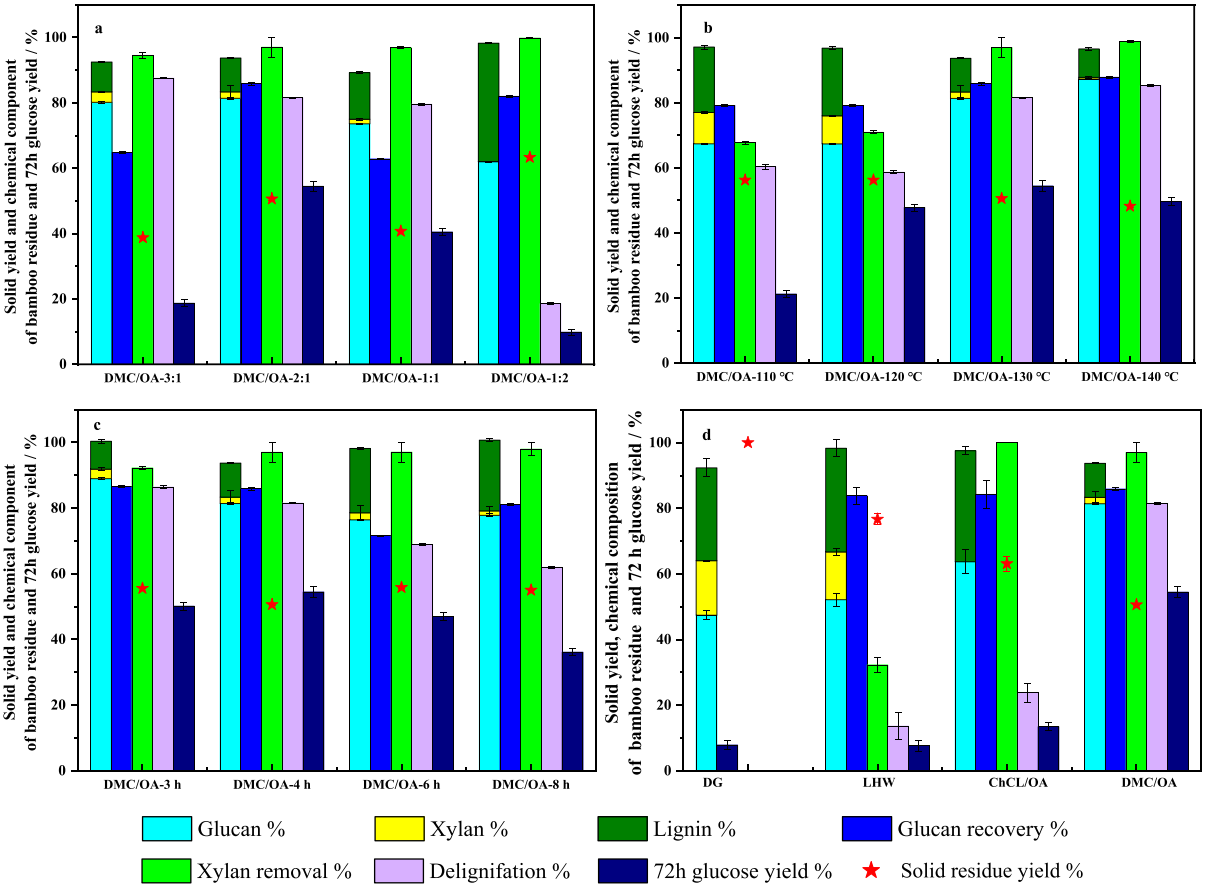


Fig. 1. The pretreatment performance of DMC/OA under different conditions. (a) Molar ratio of DMC and OA; (b) Pretreatment temperature; (c) Pretreatment time; (d) Comparison of different pretreatment methods for *D. giganteus*.

the fermentable sugar production using cellulase (Cellic® CTec2), the hydrolysis yields of 40.47%, 54.44% and 18.66% were observed. As the concentration of DMC increased, the delignification also increased, but the enzymatic hydrolysis yield showed a trend of first increasing and then decreasing, implying that there are some factors that affect the enzymatic saccharification efficiency especially under high temperature conditions, apart from the content of lignin and xylan. Besides, when the molar ratio of DMC and OA decreased to 1:2, the removal of xylan and lignin, and glucan recovery were 98.79%, 18.56% and 81.20% respectively. The subsequent enzyme hydrolysis yield was only 10.3%. This was mainly due to the fact that the high lignin content was the main factor decreasing enzyme saccharification. This suggested that more H^+ cannot damage the DG structure in DMC-based DES system, due to the proton attacking bamboo within the system had been saturated. As a result, the 2:1 molar ratio of DMC and OA was selected for DMC-based DES pretreatment.

The temperature during the DMC/OA pretreatment also greatly impacted chemical compositions in Fig. 1b. After increasing the pretreatment temperature from 110 to 140 °C, the xylan removal reached about 98%, and the delignification was increased from 59.61% to 85.34%. Meanwhile, the glucan retention slowly increased from 79.47% to 88.24%, suggesting that most of the glucan was retained in the substrate. The optimal reaction temperature was 130 °C, providing an enzymatic saccharification efficiency of 54.44%. Fig. 1c showed the effect of pretreatment time on chemical composition of bamboo. The xylan removal slightly improved, and the delignification was 86.40%, 81.53%, 68.96% and 61.95% respectively, when the time was extended from 3 to 8 h. And longer pretreatment time also can cause more glucan loss. It was noted that 4 h provided the highest enzymatic digestibility (54.44%), which was followed by 3 h (50.12%), 6 h (47.01%) and 8 h (36.12%). Therefore, 4 h was used as the appropriative pretreatment time in this study.

All the pretreatments including liquid hot water (LHW), ChCL/OA and DMC/OA DES system were conducted at 130°C for 4 h. In a controlled reaction of LHW without DES (Fig. 1d), a few amounts of lignin and xylan (19.54% and 14.53%) were removed. When bamboo powder was pretreated with DES of ChCL/OA at 130℃ for 4 h, the solid residue recovery was 63.12%, almost all xylans have been removed, and its delignification slightly increased to 23.16%. This result suggested that the ChCL/OA and DMC/OA pretreatments led to similar removal of xylan from bamboo. This may be because the dissociated protons of organic acids in DES system may break the hydrogen bond and intermolecular force between glucan and xylan, leading to the xylan removal (Ji et al., 2021). However, DMC/OA pretreatment under the optimal pretreatment conditions showed higher delignification (81.545) compared with ChCL/OA-DG (23.16%). Correspondingly, an increase of 4–7 times in saccharification efficiency of DMC/OA-DG was obtained. Because ChCL-based and DMC-based DES are both acidic DES systems, they can remove most of bamboo hemicellulose under high temperature conditions. However, there are certain differences in their delignification ability and substrate characteristics. Loow et al. (2017) and Kohli et al. (2020) had reported that Kamlet-Taft parameters in DES system and formation of hydrogen bonds between DES were critical factors affecting the dissolution of lignin and xylan.

3.2. Enhancing effect of surfactant on enzymatic hydrolysis and ethanol production

The nonionic surfactant Tween80 is used to improve the enzymatic digestibility and ethanol production of DES-treated samples. The effect of Tween80 on the release of fermentable sugars in the enzymatic saccharification of different treated samples is summarized in Fig. 2a. The addition of Tween80 (150 mg/g substrates) showed a considerable positive effect on enzymatic hydrolysis of raw DG, LHW-DG and ChCL/OA-DG. The glucose yields were 13.86% and 16.67% and 48.76%, respectively, increased by 2–3.6 folds under the Tween80-assistance. It had reported that addition of Tween 80 during the enzymatic saccharification process can reduce the non-productive cellulase binding to residual lignin and lignin-like materials on the substrates, leaving more cellulase free in liquid to hydrolyze cellulose fraction (Wang et al., 2018; Zhang et al., 2023; Zheng et al., 2020). The formation of pseudo-lignin during the ChCL/OA pretreatment of bamboo biomass could as a potential reason for inhibiting enzyme hydrolysis efficiency. Unexpectedly, the addition of Tween80 did not significantly increase the enzymatic digestibility of DMC/OA-DG, and the yield of glucose slightly increased from 54.44% to 55.18%. It is reasonable to speculate that DMC played a crucial role in preventing the deposition of pseudo-lignin formed by by-product polymerization. The mechanism may be similar that of surfactants in the enzymatic hydrolysis.

Ethanol production from hydrolysate of different pretreated substrates was conducted in a 50-mL flask through adding necessary nutrients for yeast growth, and the ethanol concentration can reveal the effects of sugars conversion. As shown in Fig. 2b, glucose levels showed a regular decreasing trend within the first 9 h, mainly because the bioconversion is influenced by the degree of enzymatic saccharification. After 12 h of fermentation, the residual glucose concentrations in all liquids fell below 0.5 g/L, suggesting that *S. cerevisiae* can effectively consumed glucose in hydrolysate and convert them into ethanol. The ethanol content reached 12.3 g/L in DMC/OA-DG, whereas the ChCL/OA-DG showed a lower ethanol production of 3.41 g/L. This is mainly because the amount of reducing sugars released from DMC/OA-DG was relatively higher. The maximum ethanol concentrations for DMC/OA-T-DG (13.5 g/L) were obtained due to the addition of Tween80 during the enzymatic hydrolysis, corresponding to the ethanol yields of 79.18%. But there was a small variation in ethanol production compared with DMC/OA -DG. Meanwhile, the ethanol concentration of ChCL/OA-T-DG

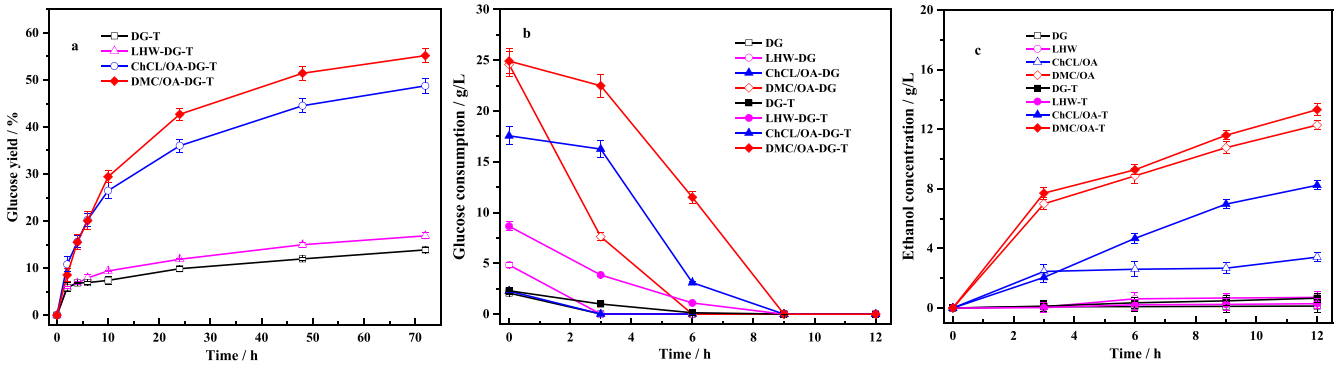


Fig. 2. Effect of nonionic surfactant Tween80 on enzymatic digestion and ethanol production of *D. giganteus*, (a) Glucose yield; (b) Glucose consumption; (c) Ethanol content.

was 8.24 g/L, which is 2.4 times higher than ChCL/OA-DG. This indicates that the addition of nonionic surfactant during the enzymatic hydrolysis obviously increased glucose production, thereby further improving the subsequent ethanol fermentation efficiency.

3.3. Effect of DMC/OA pretreatment on enzymatic adsorption and cellulose accessibility of bamboo biomass

The effects of different pretreated samples on enzymatic adsorption are shown in Fig. 3. The ability of raw DG in adsorbing cellulase was weak and its maximum amount of adsorbed enzyme (Γ_{max}), Langmuir constant (K) and distribution coefficient (R) were 13.83 mg/g sample, 4.96 mL/mg enzyme and 0.069 g/L, respectively. As shown in the adsorption isotherm, the enzyme adsorption capability on the substrates was promoted with the assistance of LHW and ChCL/OA pretreatment. Their maximum adsorption capacities were 7.08 and 18.03 mg/g sample, increased by about 24% than that of untreated DG. The affinity of cellulase to substrate (the K value) also increased to 15.29 mL/mg for LHW-DG and 17.18 mL/mg for ChCL/OA-DG. Compared with LHW, ChCL/OA pretreatment showed the higher xylan removal yield. Theoretically, the affinity of its substrate to cellulase should be stronger than LHW, because the xylan removal can result in the increase of substrates porosity. But there was no significant difference in the affinity of cellulase to the two pretreated substrates. This indicated that the surface and structural features of substrates might be the reasons for the inhibited cellulase adsorption on ChCL/OA-DG. However, when DG was performed the DMC/OA pretreatment, it was obvious that the cellulase showed higher affinity to the substrate based on adsorption curve and Langmuir constant (K), resulted in the higher adsorption capacity of cellulase on DMC/OA -DG (Γ_m=40.68 mg/g), meant that DMC/OA DES pretreatment created more binding sites of DG than the ChCL/OA pretreatment. Probable reasons for this observation were that the removal of a large amount of lignin and hemicellulose, and the absence of surface covering on the pretreated substrate surface, which will be beneficial for enzyme adsorption and hydrolysis.

The amount of cellulose surface area accessible to cellulase, defined as cellulose accessibility, also were used to predict the pretreatment performance of biomass. In Fig. 4b, the cellulose accessibility of untreated DG was only 55.46 mg dye/(g material), which is lower than that of untreated Moso bamboo (about 120 mg/g) reported by Ling et al. (2021). This was related to the content and structure of cellulose, xylan, and lignin in the raw bamboo. The value increased to 64.52 mg/g and 71.11 after LHW and ChCL/OA pretreatment, respectively. Ling et al. (2021) found that when Moso bamboo were pretreated with ChCL/OA, cellulose accessibility increased to about 135 mg/g, which was similar to the increasing trend of ChCL/OA-DG. Compared with LHW-DG, the ChCL/OA-DG was not more susceptible to enzymatic saccharification despite having most of xylan removal, and its hydrolysis yield only increased to 13.44%. This observation indicates that some covering may be on the surface of ChCL/OA-DG, causing the decrease of cellulose accessibility and enzyme digestion efficiency. Mao et al. (2010) reported that the limitations of enzymatic hydrolysis for corn stover (CS) were caused by the limited accessibility of cellulose. The DMC/OA pretreatment resulted in 3.5-fold delignification than ChCL/OA pretreatment, so the cellulose accessibility of DMC/OA-DG was further improved to 95.15 mg/g, increased by 34% than that of ChCL/OA-DG, which is in accordance with the increase of enzymatic hydrolysis efficiency. Previous studies have shown a significant positive correlation between the accessibility of cellulose and enzymatic hydrolysis (Sheng et al., 2021). The greater the accessibility of cellulose, the higher the enzymatic hydrolysis yield of substrates. However, more comprehensive research is needed to understand the impact of DMC/OA pretreatment on cellulose accessibility and enzymatic saccharification of DG bamboo, such as the structural characteristics of substrates and parameters of DMC-based DES solvent system.

3.4. The effects of DMC/OA pretreatment on the physicochemical properties of bamboo residues

The characteristics of pretreated substrates are of great significance for exploring the mechanism of DES pretreatment. FTIR spectroscopy was used to study the structural modifications of DG upon DES pretreatment. When DG was used to pretreat with DMC/OA, the peak sizes of C-O and C-O-C groups at 1032 and 1160 cm^{-1}, as well as the stretching vibration of CH_2 at 2930 cm^{-1} obviously increased in Fig. 4a, indicating an increase in relative cellulose content in the substrate. The disappearance of the two peaks at 1620 and 1511 cm^{-1} attributed to the lignin aromatic ring indicated lignin removal during the DMC/OA pretreatment process. Furthermore, the peak at around 3340 cm^{-1} of DMC/OA-DG was broader and deeper, indicating that an enhanced O-H stretching strength of hydrogen bonds in cellulose and lignin.

The hydrophobicity and degree of polymerization (DP) analyses of treated DG are shown in Fig. 4b. The DP of raw DG (1900) decreased to 693.49, and 416.19 after the DG was subjected to LHW and ChCl/OA pretreatments, respectively. The DPs of DES pretreated samples decreased by 2.7–4.6 times compared with the original bamboo, suggested that the DES system can break the glycosidic bonds in the cellulose backbone, producing short chain fibers that will facilitate enzymatic hydrolysis. In addition, the DP of DMC/OA-DG (201) showed a further downward trend, decreased by 40% compared to ChCl/OA-DG, indicating the generation of more short chain fibers. The importance of surface hydrophobicity of substrates had been also emphasized in the enzymatic adsorption and hydrolysis process (Heiss-Blanquet et al., 2011; Huang et al., 2017). Raw DG exhibits relatively high

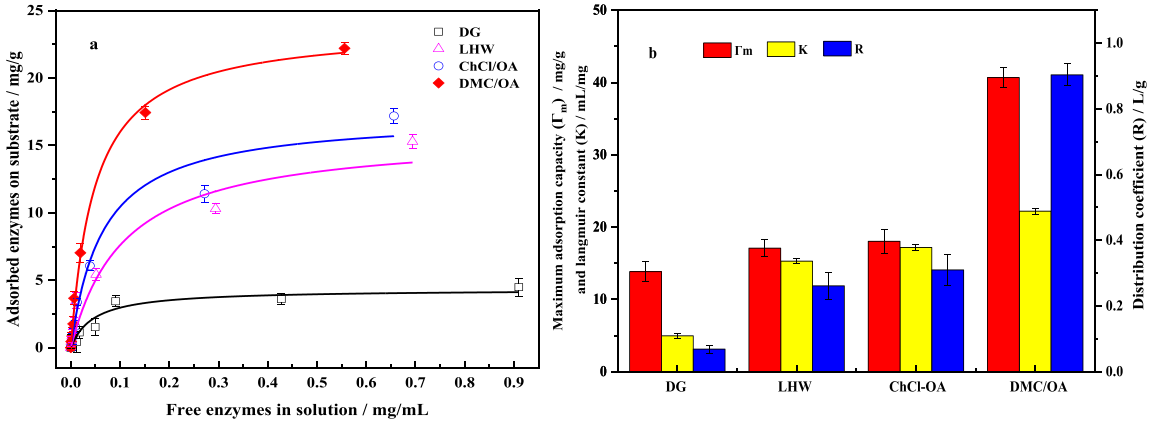


Fig. 3. Enzyme adsorption isotherms (a) and Langmuir adsorption isotherm parameters (b) on untreated and treated *D. giganteus*.

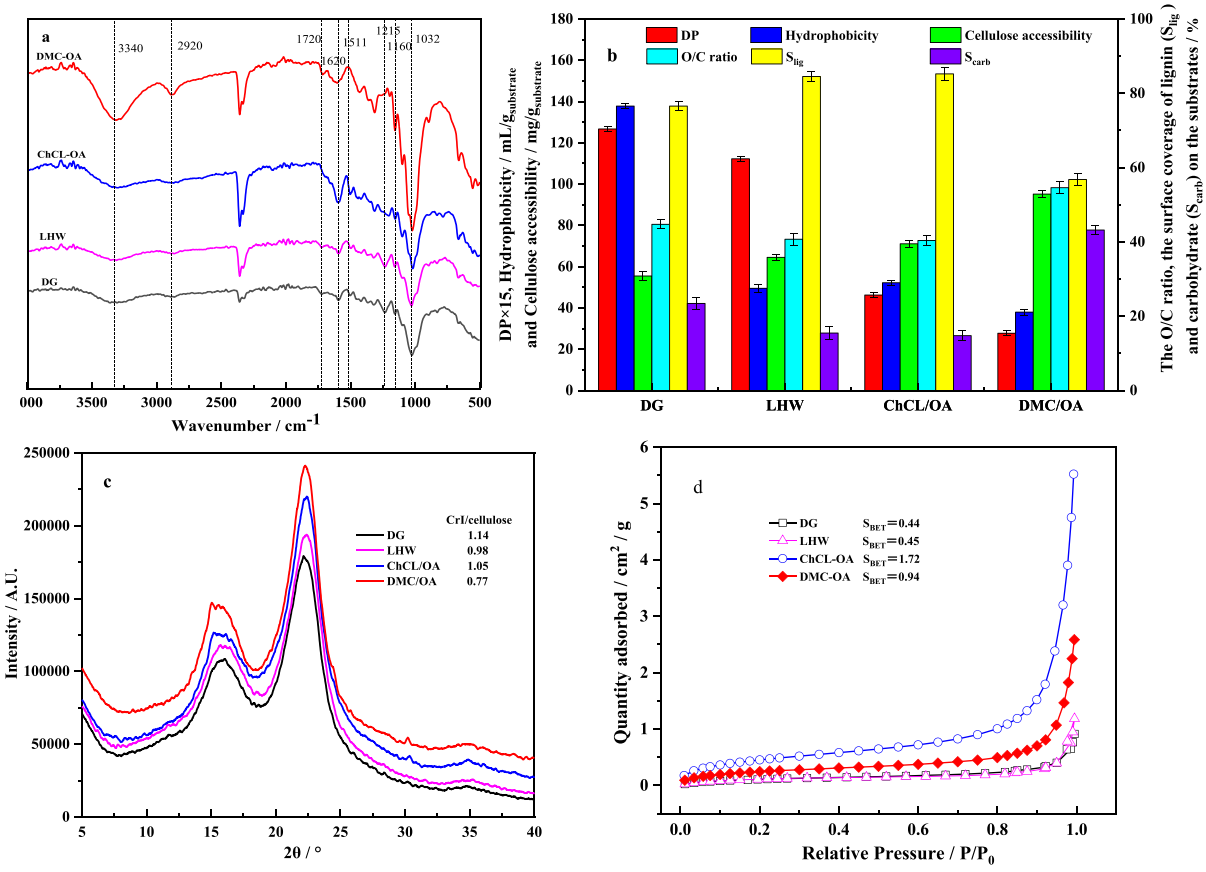


Fig. 4. The FTIR spectra (a), physical properties (b), XRD images (c) and surface area (d) of untreated and treated *D. giganteus*.

hydrophobicity of 137.8 mL/g substrate, while DMC/OA-DG (38 mL/g substrate) displayed lower hydrophobicity than ChCl/OA-DG (52.2 mL/g) and LHW-DG (49.5 mL/g). The decrease was mainly attributed to the removal of amorphous components and the increase in the relative cellulose content. The decrease of hydrophobicity can reduce the nonproductive bind of cellulase to the pretreated samples. The cellulose crystal structure of untreated and pretreated DG was determined through XRD analysis. The ratio of crystallinity to cellulose content (CrI/cellulose) can estimate the actual effect of DES pretreatments on cellulose crystallinity. The X-ray diffraction patterns of all samples showed typical cellulose I peaks in Fig. 4c. With DES pretreatment, the crystallinity of substrates decreased from 1.14 in raw DG to 1.05 in ChCl/OA-DG and 0.77 in DMC/OA-DG, respectively. The decrease in CrI/cellulose after DES pretreatment may be due to the partial removal of the xylan and lignin in the DG, and the increase in relative glucan content of substrate, which was agreed with the chemical compositions analysis in Fig. 1. These results revealed that ChCl- and DMC- based DES pretreatment were beneficial to break down the crystal structure of cellulose. It reported that there was a negative relationship between the enzyme digestion efficiency and the crystallinity of cellulose (Hall et al., 2010). As shown in Fig. 4d, the BET surface area of DG had a significant increase upon the DES pretreatment. The BET surface area of raw DG was 0.44 m^2/g. The DMC/OA pretreatment increased the surface area relative to the raw DG by 2.1 times to 0.94 m^2/g. This was attributed to depolymerization of lignin and xylan, leading to an enhance in the substrate porosity. Surprisingly, an increase of about 4-folds in the surface area of ChCl/OA-DG was observed compared with untreated DG and LHW pretreated DG. In theory, an increase of BET specific surface area can lead to an increase in cellulose accessibility, but there is no positive correlation on the two values of ChCl/OA-DG and LHW-DG. Therefore, this increase should be attributed to the formation of surface covering on the ChCl/OA-DG substrate. Although the formation of covering increases the surface area of sample, it cannot improve the cellulose accessibility due to the nonproductive adsorption between covering and enzyme. This phenomenon will also be further verified through subsequent XPS and SEM analysis.

The changes in element composition and surface chemical bonds in DG were determined through XPS analysis. as shown in Fig. 4b, the O/C ratio and coverage of surface lignin in raw DG was 44.72% and 76.57%. After LHW and ChCl-based DES pretreatment, the O/C ratio reduced to about 40%, while the coverage of surface lignin increased to about 85%. Meanwhile, the intensity of C-C and C-H bonds (284.6, eV) represented the lignin content was slightly higher as compare to untreated DG, indicating that the surface of LHW-DG and ChCl/OA-DG had a higher concentration of lignin and the surface covering could come from the degradation products of lignin and xylan (such as pseudo lignin and surface lignin). Interestingly, even higher O/C ratio (54.6%) and lower C-C and C-H peak intensity were found in DMC/OA-DG, suggested that DMC-based DES pretreatment can further decrease the lignin content. In addition, the peaks intensities of C-O, O-C═O and C═O belonged to carbohydrate in DMC/OA-DG were higher than that of ChCl/OA-DG and LHW-DG (Fig. 5), which was consistent with the content of surface carbohydrates in the DMC/OA-DG (22.6%), ChCl/OA-DG (14.78%) and LHW-DG (15.46%). This indicated that the ratio of cellulose on the bamboo surface was increased, and some cellulose components were exposed, which will facilitate the enzymatic digestion of cellulose. The result revealed that DMC/OA pretreatment had good abilities to inhibit the formation of surface lignin and pseudo lignin.

By comparing the SEM images (Fig. 6), the morphological changes and surface damage of bamboo prior to and after pretreatment were studied. The surface of untreated DG was much flat, smooth and compact, but small structural disintegrations due to size reduction and

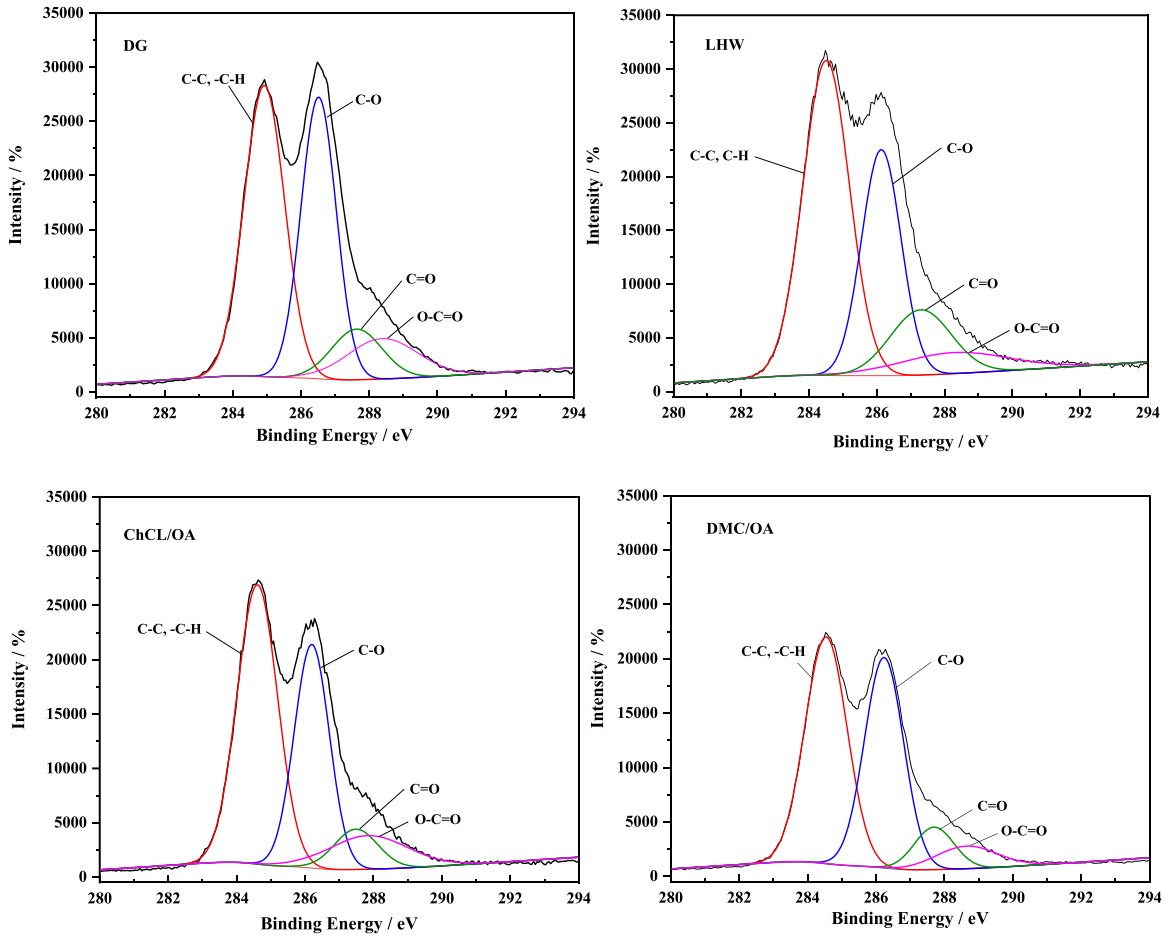


Fig. 5. XPS analysis of untreated and pretreated *D. giganteus*.

milling. After ChCl/OA pretreatment, the length and width of the material decreased, and cracks appeared on the surface of the sample. And a large amount of lignin particles was observed on the surface of ChCl/OA-DG due to the lignin migration. There were not the lignin droplets on the SEM images of LHW-DG, which may be due to the pretreatment temperature of 130 °C cannot allow lignin to migrate to the substrate surface. In the DMC/OA DES system, the obtained substrates became rough, loose and irregular with numerous cracks, holes, and wrinkles. And surprisingly, the disappearance of lignin droplets was observed in the SEM images of DMC/OA-DG. The successful removal of surface lignin was also proved by the decrease trend of C-C and C-H peak intensity in XPS spectra. Generally speaking, the biomass pretreatment at higher temperatures, especially acidic condition can produce pseudo lignin that can nonspecific adsorb cellulase, resulted in the reduce of enzymatic hydrolysis efficiency (Hu et al., 2012). The addition of surfactants and bovine serum albumin can relieve the inhibitory effect on the enzymatic hydrolysis, but they are expensive and not suitable for large-scale use in the biomass biorefinery. It was obvious that the DMC/OA solvent system had additional functions similar to surfactants, and can prevent lignin droplets from approaching the substrate or their formation, indicating that the DMC-based DES systems with self-cleaning lignin droplets have an excellent ability on the biomass pretreatment.

3.5. Molecular weight and structural characteristics of DMC/OA-lignin fractions

The molecular weight of original lignin (MWL) and DMC/OA-lignin fractions were calculated from GPC curves and compared (see supplementary material). The Mw and Mn of original lignin (MWL-DG) were about 3484 g/mol and 1357 g/mol, while DMC/OA-lignin exhibited low weight-average molecular weights, 1527 g/mol and 731 g/mol, respectively, suggesting that the significantly depolymerization of lignin structure took place during the DMC/OA pretreatment. Zhou et al. (2024) found that the Mw of lignin from the DES pretreatment with ChCl-LA was about 900–1300 g/mol. In addition, it was found that DMC/OA pretreatment was beneficial for improving the uniformity of lignin molecular weight, and the PDI value reduced from 2.57 to 2.08 after the DES pretreatment. The DES lignin had relatively low molecular weight and a uniform molecular structure, which have great potential in the practical biorefinery process.

The structural variations of lignin from the DMC/OA pretreatment were investigated by 2D-HSQC NMR technique (Ma et al., 2021; Fu et al., 2024). The MWL and DMC/OA-lignin have strong signals at δ_C/δ_H 56.2/3.72 ppm on the region of side chain region, indicating that there was a large of methoxy groups on the substrates. Compared to MWL, the signal of DMC/OA-lignin was weak in A_γ (δ_C/δ_H 59.4/3.62 ppm), A'_γ (δ_C/δ_H 62.8/4.29 ppm) and Aβ (δ_C/δ_H 84.0/4.28 ppm), suggesting that the scission of β-O-4 aryl ethers was the major mechanism of lignin breakdown in the DMC-based DES pretreatment. The increase in the signal of B_β (C_α–H_α in β-5, δ_C/δ_H 53.6/3.04) indicated more condensation reactions occurred at the C_β position due to high temperature and the presence of oxalic acid. In addition, the aromatic region of DMC/OA lignin only retains the basic syringol (S) and hydroxyphenyl (H) skeleton structures, while the other signals disappeared (see supplementary material). This fact also reflected that both depolymerization and condensation of lignin occurred during the DES pretreatment.

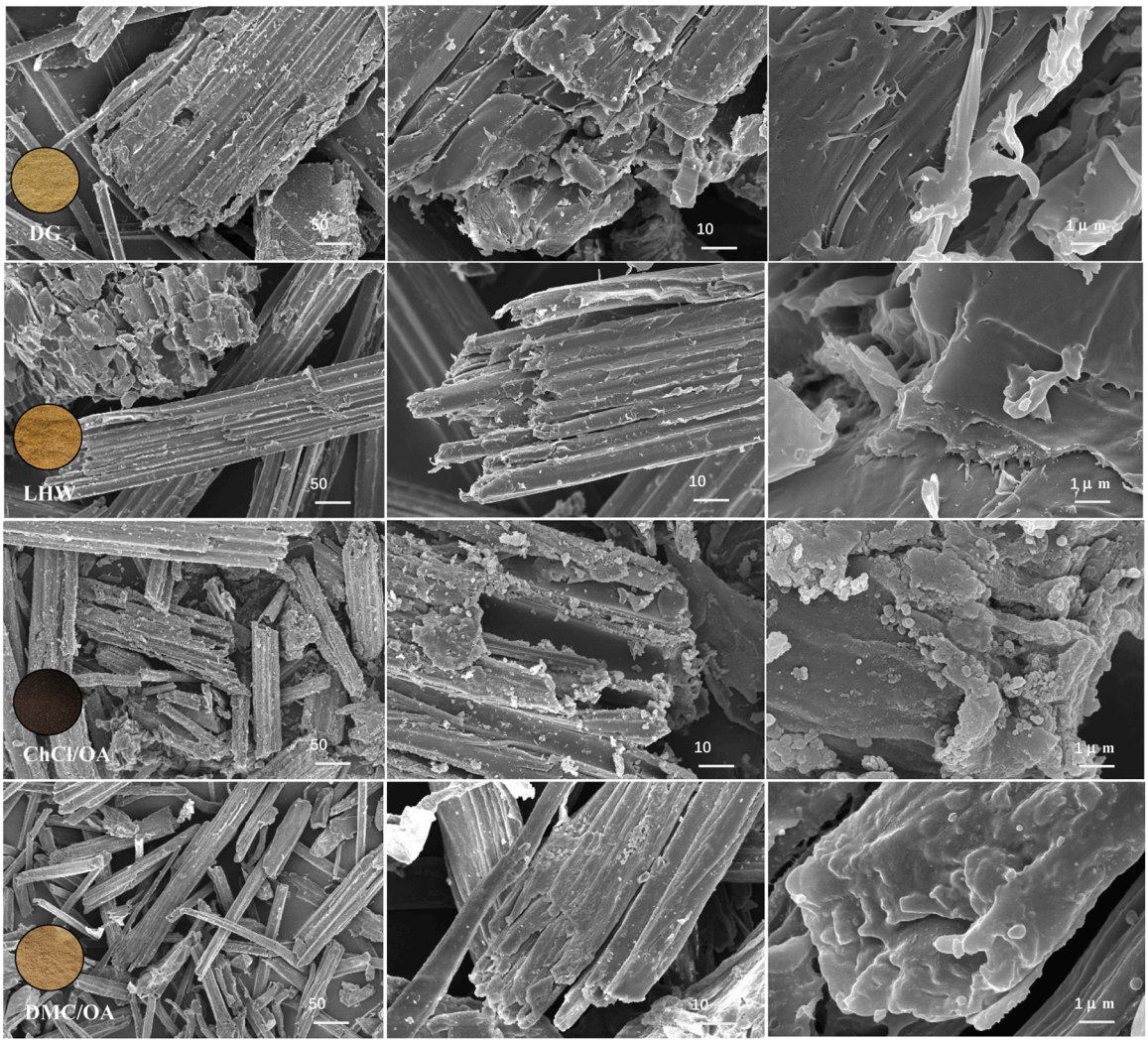


Fig. 6. SEM micrographs of *D. giganteus* using different pretreatment.

3.6. Properties of DES and its interaction with cellulose-hemicellulose-lignin systems

The Kamlet-Taft empirical parameters (α, β, and π*) of ChCl/OA and DMC/OA solvents were shown in Fig. 7a. The values of α, β, and π* represented respectively the ability to provide and accept hydrogen bonds, as well as the polarizability of solvents, which can be affected by the structures of DES system. Both ChCl and DMC are quaternary ammonium salts, but the alkyl chain length of DMC is longer than that of ChCl, which can affect the Kamlet-Taft parameters of DES system. As can be seen in Fig. 7a, the increase of α and π* value in ChCl/OA (1.802 and 1.408) was about 6% compared to these of DMC/OA. The α and π* values of two DES systems showed a slight difference due to the similar HBD. Liu et al. (2019) and Yu et al. (2020) reported that the lignin solubility had a positive correlation with the α and π* of DESs, while the xylan solubility had a positive correlation with β. This meant that DES of ChCl/OA should have better lignin solubility than DMC/OA. In addition, it was observed that there was an increasing trend in the β value for ChCl/OA (0.1032) and DMC/OA (0.3678). It indicates that DMC had strong hydrogen bond acceptance ability, thus the DMC/OA solvent system could be a strong ability to break intermolecular hydrogen bonds in biomass, which could lead to the decrease of DP in the cellulose substrate (Fig. 4b). Unfortunately, the hydrogen bonds interactions heightened the viscosity of DMC-based DES. The viscosity DMC/OA (25.41 mPa·s) was higher than the ChCl/OA (15.65 mPa·s). Hernandez et al. (2013) had reported that viscosity below 90 mPa·s was beneficial for the fluidity and mass transfer of substrates in the pretreatment process.

The conversion of hydrogen bonds interactions was also evaluated by classic molecular dynamics simulations. The cellulose-hemicellulose-lignin system without DES, as a control group, was simulated. As depicted in Fig. 7b, the average number of hydrogen bonds between cellulose-hemicellulose-lignin system was 11989, meant that the model system had strong hydrogen bonding. However, when DES was added to the cellulose-xylan-lignin system, the hydrogen bonding network of the original system underwent significant changes. The addition of ChCl/OA and DMC/OA can result in the decrease of 33.4 and 26.1 times in the number of hydrogen bonds of the original system. The addition of DES can generate new hydrogen bonds with cellulose and xylan. Although the amount of hydrogen bonds with lignin was relatively less, DES can still effectively break the interaction between cellulose, hemicellulose, and lignin. The xylan-glucan, glucan-lignin and xylan-lignin system exhibited higher interaction strengths including electrostatic interactions and van der Waals forces, with total absolute values of 2.37×10^6, 1.82×10^6 and 1.28×10^6 respectively in the Table 1. It was observed that the highest absolute values of interaction energy decreased to 71756 kJ/mol, 53500 kJ/mol and 40008 kJ/mol for the addition of ChCl/OA, 98826 kJ/mol, 73492 kJ/mol and 54818 kJ/mol for the addition of DMC/OA, respectively, when DES solvents were introduced to the cellulose-xylan-lignin system. The results showed that

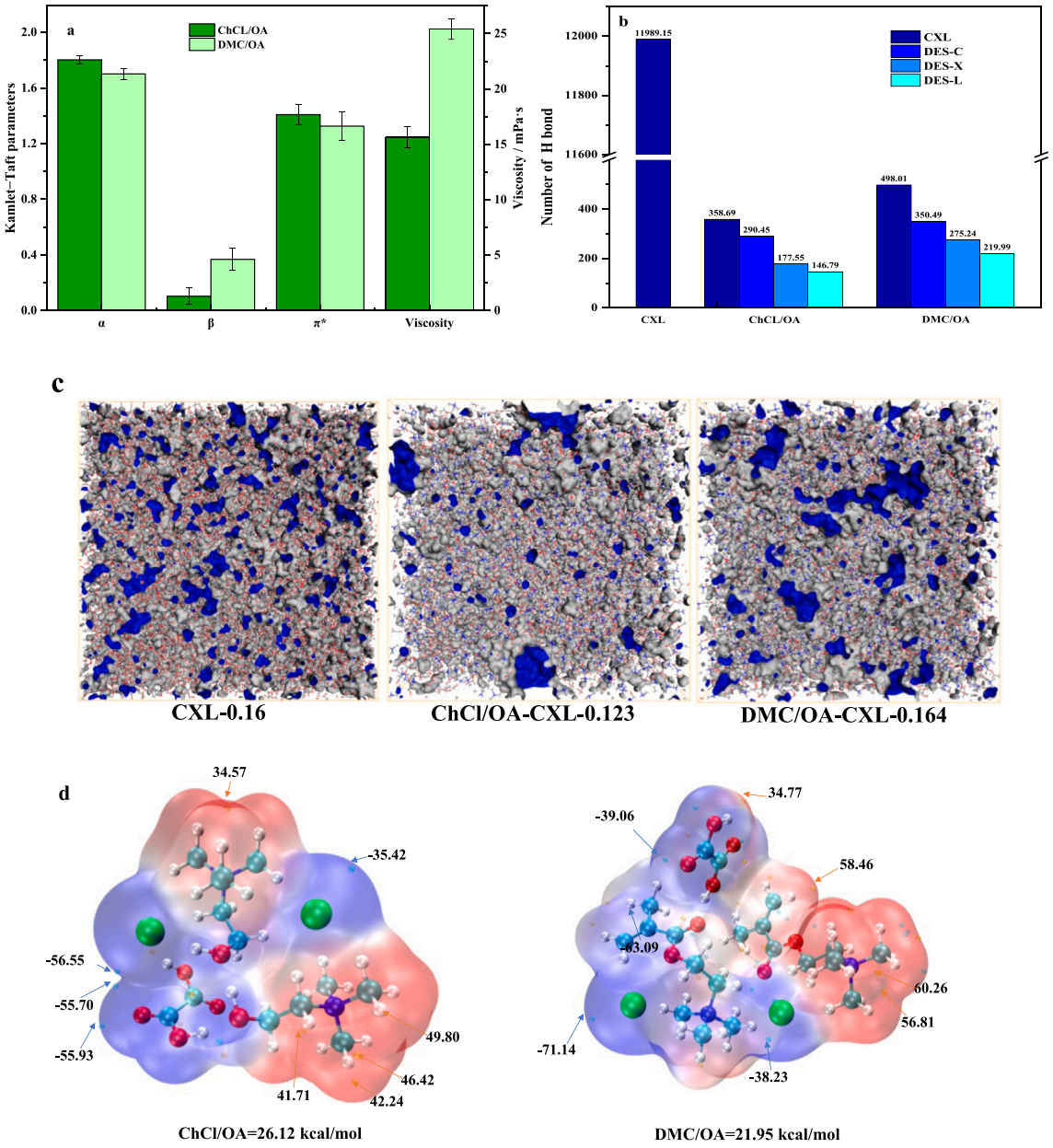


Fig. 7. The Kamlet-Taft empirical parameters of DMC- and ChCl-based DES (a); Average number of hydrogen bonds (b) and free volume fraction (c) during the Cellobiose-4-O-methyl gluconic acid xylose-Veratrylglycerol-β-guaiacyl ether system; The geometric configuration and electrostatic potential surface of DMC/OA and ChCl/OA (d).

the addition of DES can significantly decrease the interaction energy and further destroy the hydrogen bond network structure of the original model complexes, which facilitate the separation and dissolution of biomass components. Meanwhile, free volume was used to determine the tightness of molecular bonding in the solvent system. The smaller the free volume fraction, the closer the molecules fit together in the system. The cellulose-xylan-lignin system showed the free volume fraction of 0.16 in Fig. 7c. The addition of ChCl/OA reduced the free volume fraction to 0.123, while DMC/OA led to a slight high free volume fraction (0.164). There was a certain relationship between the free volume fraction and the viscosity of the solution.

It was observed that the hydrogen bonds number and the interaction energy in the ChCl/OA and model compound system was slightly lower than the DMC/OA system, suggesting that the lignin dissolution in the ChCl/OA should be better than that in the DMC/OA system. However, our experimental results (Fig. 1) are slightly inconsistent with the theoretical results. This was mainly because the model compound (cellobiose, 4-O-methyl gluconic acid xylose and veratrylglycerol-β-guaiacyl ether) used are much smaller than the natural cellulose, hemicellulose and lignin in the lignocellulosic biomass. Free DMC/OA and ChCl/OA without hydrogen bonding with model compound can also participate in the chemical calculation process, and the DMC has a longer the alkyl chain length than ChCl, which will lead to the generation of more hydrogen bonds and interaction energy in the system. Furthermore, the presence of lignin droplets on the surface of substrate also can lead to its high total lignin content during the ChCl/OA pretreatment. In addition, the ratio of HBA to HBD in ChCl/OA (2:1) was not the optimal molar ratio, which also could one of the reasons for poor pretreatment performance. To further explore the differences between these two DES systems, the geometric configuration and electrostatic potential surface of the ChCl/OA and DMC/OA systems were shown in Fig. 7d, where the blue part represents negative electrostatic potential,

Table 1
Interaction energy of Cellobiose-4-O-methyl gluconic acid xylose-Veratrylglycerol-β-guaiacyl ether system with the addition of DES.

Model System	Interaction Energy / kJ/mol	CHL	ChCl/OA	DMC/OA
Cellobiose/4-O-methyl gluconic acid xylose	E_{Coul}	-2.28×10^{6}	-7.08×10^{4}	-9.78×10^{4}
	E_{vwd}	-9.04×10^{4}	-9.56×10^{2}	-1.03×10^{3}
	E_{total}	-2.37×10^{6}	-7.18×10^{4}	-9.89×10^{4}
Cellobiose/Veratrylglycerol-β-guaiacyl ether	E_{Coul}	-1.68×10^{6}	-5.23×10^{4}	-7.22×10^{4}
	E_{vwd}	-1.42×10^{5}	-1.20×10^{3}	-1.26×10^{3}
	E_{total}	-1.82×10^{6}	-5.35×10^{4}	-7.35×10^{4}
4-O-methyl gluconic acid xylose/Veratrylglycerol-β-guaiacyl ether	E_{Coul}	-1.21×10^{6}	-3.94×10^{4}	-5.41×10^{4}
	E_{vwd}	-6.74×10^{4}	-6.45×10^{2}	-7.53×10^{2}
	E_{total}	-1.28×10^{6}	-4.00×10^{4}	-5.48×10^{4}

the white part zero electrostatic potential, and the red part positive electrostatic potential, respectively. Although the molecular polarity index of ChCl-based DES (26.12 kcal/mol) is slightly higher than that of DMC-based DES (21.95 kcal/mol), the higher negative electrostatic potential has been observed on DMC/OA system, which may be more favorable for the interaction between DES and lignocellulosic biomass, and finally lead to the dissolution of the amorphous region of biomass.

From the above discussion, multi-effects were revealed in the DMC/OA pretreatment of bamboo residues. The DES system of DMC/OA with a long the alkyl chain length, had strong hydrogen bond acceptance ability and can destroy the hydrogen bond network structure of bamboo material. And the DMC/OA can not only catalyze the cleavage of the ether or ester bond between lignin and hemicellulose, but disrupted part of cellulose glycosidic bonds, ultimately leading to a decrease in cellulose polymerization and crystallinity degree. More importantly, the DMC/OA pretreatment has a significant self-cleaning effect on pseudo-lignin and surface lignin on the substrates. Additionally, due to the dissolution of lignin and xylan in the DMC-based DES system, the changes of hydrophilicity/hydrophobicity properties, crystal structures of cellulose, and cellulose accessibility were beneficial for the enzymatic saccharification. A new type of DMC/OA treated bamboo residue with self-cleaning function has been synthesized, which can clean the lignin droplets formed on the surface of substrate during the pretreatment process, and ultimately improve the enzymatic saccharification efficiency.

3.7. Mass balance analysis of ChCl/OA and DMC/OA pretreatment

The pretreatment, enzymatic hydrolysis and ethanol fermentation process of bamboo wastes are described in the mass balance image (Fig. 8). The 1000 g of DG including 474.7 g glucan, 166.1 g xylan and 283.1 g lignin were subjected to the DES pretreatment at a solid-liquid ratio of 1:10 at 130 °C for 4 h. When ChCl/OA was used as pretreatment solvent, 780.3 g of residue was recovered, which consisted of 402.1 g glucan, 1.2 g xylan and 213.7 g lignin. Then 54 g of glucose and a low ethanol content of 3.41 g/L (corresponding to ethanol yield of 61.18%) was obtained during the enzymatic hydrolysis and fermentation of ChCl/OA -DG. However, after DMC/OA pretreatment, 506.1 g pretreated DG were achieved, containing glucan of 411.9 g, xylan of 9.9 g and lignin of 52.4 g, indicating a large amount of lignin and hemicellulose have been removed, while most of the glucan was retained. Glucose of 224.2 g was released from the enzymatic hydrolysis process of DMC/OA-DG, with a glucose yield of 54.44%. The fermentation of hydrolysates resulted in the maximum ethanol production of 12.3 g/L (corresponding to ethanol yield of 73.25%), which were 3.6-folds higher compared with that of ChCl/OA -DG.

Previous studies based on various bamboo pretreatment were summarized (see supplementary material). It showed that the DMC-based DES pretreatment performance was at a moderate level among all bamboo pretreatments. The highest enzymatic hydrolysis yield (about 65%) comes from ternary DES pretreatment (ChCl/OA/EG), which is 1.18 times higher than DMC/OA pretreatment. During the pretreatment of ChCl/OA/EG, a third component, ethylene glycol, was required to alleviating the surface lignin and pseudo lignin on the substrates. DES of DMC/OA has similar to the effect of ternary DES pretreatment. Additionally, most methods improved the pretreatment performance of bamboo using the diverse hydrogen donors, while DMC, as hydrogen receptor, has the advantages of low cost, low energy consumption, and ease of preparation, which DMC/OA pretreatment has potential in improving biorefinery efficiency of bamboo and reducing costs.

4. Conclusions

A new green solvent DMC/OA with the self-cleaning lignin droplets function was used to treat bamboo at 130 °C and a solid-liquid ratio of 2:1 for 4 h. It removed a large amount of lignin and xylan from biomass, and the coverage of surface lignin was reduced to 56%, which increased the enzymatic digestibility by 4 times compared to ChCl-based DES. Higher cellulose accessibility, lower degree of polymerization, hydrophobicity and CrI/cellulose, and more porous structure of DMC/OA-DG were obtained. It provides some useful information for developing the functional DES pretreatment and more green solvents for the bamboo biorefinery.

CRediT authorship contribution statement

Dawei Wang: Visualization, Validation, Data curation. **Haiyan Yang:** Validation, Software. **Zhengjun Shi:** Resources, Investigation. **Zhaomin Liu:** Formal analysis, Data curation. **Qinpei Hou:** Writing – original draft, Methodology. **Jing Yang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

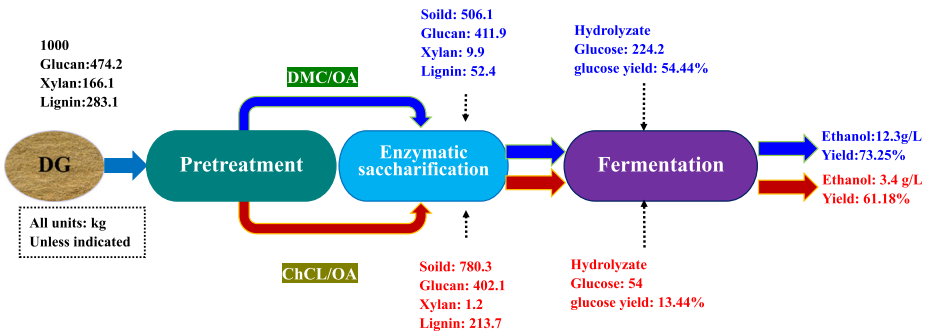


Fig. 8. Mass balance of DMC/OA pretreatment, enzymatic saccharification and fermentation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

Acknowledgments

The authors would like to express their gratitude for the National Natural Science Foundation of China (No. 32060329), Yunnan Province Agricultural Basic Research Joint Special Project (No. 202301BD070001–157), Reserve Talent for Young and Middle aged Academic, Technical Leaders of Yunnan Province (202005AC160044) and Xing Dian Youth Talents Support Program of Yunnan Province (XDYC-QNRC-2022–0175).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2024.118730.

References

Ahmed, S.F., Mofijur, M., Chowdhury, S.N., Nahrin, M., Rafa, N., Chowdhury, A.T., Nuzhat, S., Ong, H.C., 2022. Pathways of lignocellulosic biomass deconstruction for biofuel and value-added products production. Fuel 318, 123618. https://doi.org/10.1016/j.fuel.2022.123618.

Areeya, S., Panakkal, E.J., Sriatiyanun, M., Kangsadan, T., Tawai, A., 2023. A review on chemical pretreatment of lignocellulosic biomass for the production of bioproducts: mechanisms, challenges and applications. Appl. Sci. Eng. Prog. 16 (3), 6767. https://doi.org/10.14416/j.asep.2023.02.008.

Chen, Z., Bai, X., A., L., Zhang, H., Wan, C., 2020. Insights into structural changes of lignin toward tailored properties during deep eutectic solvent pretreatment. ACS Sustain. Chem. Eng. 8 (26), 9783–9793. https://doi.org/10.1021/acssuschemeng.0c01361.

Deng, H.Y., Xu, Y.Y., Chen, Q.C., Wei, X.Z., Zhu, B.K., 2011. High flux positively charged nanofiltration membranes prepared by UV-initiated graft polymerization of methacrylatoethyl trimethyl ammonium chloride (DMC) onto polysulfone membranes. J. Membr. Sci. 366, 363–372. https://doi.org/10.1016/j.memsci.2010.10.029.

Frank, N., 2022. Software update: The ORCA program system-Version 5.0. WIREs Comput. Mol. Sci. 12 (5), 1606. https://doi.org/10.1002/wcms.1606.

Fu, C., Wang, S.Z., Song, G.Y., 2024. Efficient fractionation of corn stover using deep eutectic solvent from lignin-derived 3-phenylpropionic acid: towards a closed-loop biorefinery. Ind. Crop. Prod. 212, 118380 https://doi.org/10.1016/j.indcrop.2024.118380.

Gessner, A., Waicz, R., Lieske, A., Paulke, B.R., Mäder, K., Müller, R.H., 2000. Nanoparticles with decreasing surface hydrophobicities: influence on plasma protein adsorption. Int. J. Pharm. 196 (2), 245–249. https://doi.org/10.1016/S0378-5173(99)00432-9.

Hall, M., Bansal, P., Lee, J.H., Realff, M.J., Bommarius, A.S., 2010. Cellulose crystallinity-a key predictor of the enzymatic hydrolysis rate, 277 FEBS J. 619 (6), 1571–1582. https://doi.org/10.1111/j.1742-4658.2010.07585.

Heiss-Blanquet, S., Zheng, D., Ferreira, N.L., Lapierre, C., Baumberger, S., 2011. Effect of pretreatment and enzymatic hydrolysis of wheat straw on cell wall composition, hydrophobicity and cellulase adsorption, 630 Bioresour. Technol. 102 (10), 5938–5946. https://doi.org/10.1016/j.biortech.2011.03.011.

Hernandez, E., Garcia, A., Lopez, M., Puls, J., Parajo, J.C., Martin, C., 2013. Dilute sulphuric acid pretreatment and enzymatic hydrolysis of Moringa oleifera empty pods. Ind. Crop. Prod. 44, 227–231. https://doi.org/10.1016/j.indcrop.2012.11.001.

Hu, Fan, Jung, S.W., Ragauskas, A., 2012. Pseudo-lignin formation and its impact on enzymatic hydrolysis. Bioresour. Technol. 117 (2012), 7–12. https://doi.org/10.1016/j.biortech.2012.04.037.

Huang, Y., Sun, S., Huang, C., Yong, Q., Elder, T., Tu, M., 2017. Stimulation and inhibition of enzymatic hydrolysis by organosolv lignins as determined by zeta potential and hydrophobicity. Biotechnol. biofuels 10 (1), 162. https://doi.org/10.1186/s13068-017-0853-6.

Humphrey, W., Dalke, A., Schulten, K., 1996. VMD-visual molecular dynamics. J. Molec. Graph. 14 (1), 33–38. https://doi.org/10.1016/0263-7855(96)00018-5.

Inglesby, M.K., Zeronian, S.H., 2002. Direct dyes as molecular sensors to characterize. Cellul. Substr. Cellul. 9, 19–29. https://doi.org/10.1023/A:1015840111614.

Ji, Q.H., Tan, C.P., Yagoub, A.E.A., Chen, L., Yan, D., Zhou, C.S., 2021. Effects of acidic deep eutectic solvent pretreatment on sugarcane bagasse for efficient 5-hydroxymethylfurfural production. Energy Technol. 9, 2100396 https://doi.org/10.1002/ente.202100396.

Jose, D., Tawai, A., Divakaran, D., Bhattacharyya, D., Venkatachalam, P., Tantayotai, P., Sriariyanun, M., 2023. Integration of deep eutectic solvent in biorefining process of lignocellulosic biomass valorization. Bioresource Technology Reports 21, 101365. https://doi.org/10.1016/j.biteb.2023.101365.

Kohli, K., Katuwal, S., Biswas, A., Sharma, B.K., 2020. Effective delignification of lignocellulosic biomass by microwave assisted deep eutectic solvents. Bioresour. Technol. 303, 122897 https://doi.org/10.1016/j.biortech.2020.122897.

Li, A.L., Hou, X.D., Lin, K.P., Zhang, X., Fu, M.H., 2018. Rice straw pretreatment using deep eutectic solvents with different constituents molar ratios: biomass fractionation, polysaccharides enzymatic digestion and solvent reuse. J. Biosci. Bioeng. 126, 346–354. https://doi.org/10.1016/j.jbiosc.2018.03.011.

Liang, Z.W., Neményi, A., Kovács, G.P., Gyuricza, C., 2023. Potential use of bamboo resources in energy value-added conversion technology and energy systems. GCB Bioenergy 15, 936–953. https://doi.org/10.1111/gcbb.13072.

Lin, W.Q., Yang, J.L., Zheng, Y.Y., Huang, C.X., Yong, Q., 2021. Understanding the effects of different residual lignin fractions in acid-pretreated bamboo residues on its enzymatic digestibility. Biotechnol. Biofuels 14, 143. https://doi.org/10.1186/s13068-021-01994-y.

Ling, Z., Tang, W., Su, Y., Shao, L.P., Wang, P., Ren, Y.X., Huang, C.X., Lai, C.H., Yong, Q., 2021. Promoting enzymatic hydrolysis of aggregated bamboo crystalline cellulose by fast microwave-assisted dicarboxylic acid deep eutectic solvents pretreatments. Bioresour. Technol. 333 (2021), 125122 https://doi.org/10.1016/j.biortech.2021.125122.

Liu, Q.L., Mou, H.Y., Chen, W.J., Zhao, X.H., Yu, H.T., Xue, Z.M., Mu, T.C., 2019. Highly efficient dissolution of lignin by eutectic molecular liquids. Ind. Eng. Chem. Res. 58, 23438–23444. https://doi.org/10.1021/acs.iecr.9b05059.

Liu, Y.Z., Chen, W.S., Xia, Q.Q., Guo, B.T., Wang, Q.W., Liu, S.X., Liu, Y.X., Li, J., Yu, H. P., 2017. Efficient cleavage of lignin-carbohydrate complexes and ultrafast extraction of lignin oligomers from wood biomass by microwave-assisted treatment with deep eutectic solvent. Chem. SusChem. 10, 1692–1700. https://doi.org/10.1002/cssc.201601795.

Loow, Y.L., New, E.K., Yang, G.H., Ang, L.Y., Foo, L.Y.W., Wu, T.Y., 2017. Potential use of deep eutectic solvents to facilitate lignocellulosic biomass utilization and conversion. Cellulose 24 (9), 3591. https://doi.org/10.1007/s10570-017-1358-y.

Lu, T., Chen, F., 2012. Multiwfn: a multifunctional wavefunction analyzer. J. Comput. Chem. 33 (5), 580–592. https://doi.org/10.1002/jcc.22885.

Lugani, Y., Rai, R., Prabhu, A.A., Maan, P., Hans, M., Kumar, V., Kumar, S., Chandel, A. K., Sengar, R.S., 2020. Recent advances in bioethanol production from lignocelluloses: a comprehensive review with a focus on enzyme engineering and designer biocatalysts. Biofuel Res. J. 7, 1267–1295. https://doi.org/10.18331/BRJ2020.7.4.5.

Ma, C.Y., Gao, X., Peng, X.P., Gao, Y.F., Liu, J., Wen, J.L., Yuan, T.Q., 2021. Microwave-assisted deep eutectic solvents (DES) pretreatment of control and transgenic poplars for boosting the lignin valorization and cellulose bioconversion. Ind. Crop. Prod. 164, 113415 https://doi.org/10.1016/j.indcrop.2021.113415.

Mao, J.D., Holtman, K.M., Franqui-Villanueva, D., 2010. Chemical structures of corn stover and its residue after dilute acid prehydrolysis and enzymatic hydrolysis: insight into factors limiting enzymatic hydrolysis. J. Agric. Food Chem. 58 (22), 11680–11687. https://doi.org/10.1021/jf102514r.

Meng, F.Y., Fan, J., Cui, F., Yang, H.Y., Shi, Z.J., Wang, D.W., Yang, J., 2023. An innovative and efficient hydrogen peroxide-citric acid pretreatment of bamboo residues to enhance enzymatic hydrolysis and ethanol production. Bioresour. Technol. 383, 12923. https://doi.org/10.1016/j.biortech.2023.129230.

Pihlajaniemi, V., Sipponen, M.H., Liimatainen, H., Sirviö, J.A., Nyyssölä, A., Laakso, S., 2016. Weighing the factors behind enzymatic hydrolyzability of pretreated lignocellulose. Green. Chem. 18 (5), 1295–1305. https://doi.org/10.1039/C5GC01861G.

Scurlock, J.M.O., Dayton, D.C., Hames, B., 2000. Bamboo: an overlooked biomass resource? Biomass-.-. Bioenerg. 19, 229–244. https://doi.org/10.1016/S0961-9534(00)00038-6.

Selkälä, T., Sirviö, J.A., Lorite, G.S., Liimatainen, H., 2016. Anionically stabilized cellulose nanofibrils through succinylation pretreatment in urea-lithium chloride deep eutectic solvent. ChemSusChem 9, 3074–3083. https://doi.org/10.1002/cssc.201600903.

Shen, X.J., Chen, T., Wang, H.M., Mei, Q., Yue, F., Sun, S., Wen, J.L., Yuan, T.Q., Sun, R. C., 2020. Structural and morphological transformations of lignin macromolecules during bio-based deep eutectic solvent (DES) pretreatment. ACS Sustain. Chem. Eng. 8 (5), 2130–2137. https://doi.org/10.1021/acssuschemeng.9b05106.

Sheng, Y.Q., Tan, X., Gu, Y.J., Zhou, X., Tu, M.B., Xu, Y., 2021. Effect of ascorbic acid assisted dilute acid pretreatment on lignin removal and enzyme digestibility of agricultural residues. Renew. Energy 163, 732–739. https://doi.org/10.1016/j.renene.2020.08.135.

Sluiter, A., Hames, B., Ruiz, R., Scarlata, C., Sluiter, J., Templeton, D., Crocker, D., 2011. Determination of Structural Carbohydrates and Lignin in Biomass, Laboratory Analytical Procedure (LAP) NREL/TP-510−42618. National Renewable Energy Laboratory, Golden CO.

Smink, D., Juan, A., Schuur, B., Kersten, S.R.A., 2019. Understanding the role of choline chloride in deep eutectic solvents used for biomass delignification, 2019 Ind. Eng. Chem. Res. 58, 16348–16357. https://doi.org/10.1021/acs.iecr.9b03588.

Tang, Z.Y., Wu, C.Q., Tang, W., Ma, C.L., He, Y.C., 2023. A novel cetyltrimethylammonium bromide-based deep eutectic solvent pretreatment of rice husk to efficiently enhance its enzymatic hydrolysis. Bioresour. Technol. 376, 128806 https://doi.org/10.1016/j.biortech.2023.128806.

Wang, C.G., Yang, S.Q., Song, X.B., Pi, Q.F., Zhang, Q., Liu, Q.Y., Xu, Y., Chen, L.G., Ma, L.L., 2020. Novel solvent systems for biomass fractionation based on hydrogen-bond interaction: a minireview. Adv. Sustain. Syst. 4 (10), 2000085 https://doi.org/10.1002/adsu.202000085.

Wang, W., Zhuang, X.S., Tan, X.S., Wang, Q., Chen, X.Y., Yu, Q., Qi, W., Wang, Z.M., Yuan, Z.H., 2018. Dual effect of nonionic surfactants on improving the enzymatic hydrolysis of lignocellulose. Energ. Fuel. 32, 5951–5959. https://doi.org/10.1021/acs.energyfuels.8b00225.

Wang, Y.H., Zhang, X.R., Li, W.J., Cheng, J.L., Liu, C.C., Zheng, J.J., 2016. Determination of reactivity ratios of copolymerization of acrylamide (AM) and methacryloxyethyltrimethyl ammonium chloride (DMC) with ultraviolet initiation, and their sequence length distribution. Polym. Polym. Compos. 24 (5), 307–314. https://doi.org/10.1177/096739111602400501.

Xue, Z., Zhao, X., Sun, R., Mu, T., 2016. Biomass-derived γ-valerolactone-based solvent systems for highly efficient dissolution of various lignins: dissolution behavior and mechanism study. ACS Sustain. Chem. Eng. 4 (7), 3864–3870. https://doi.org/10.1021/acssuschemeng.6b00639.

Yang, L., Xu, L.Q., Yang, H.Y., Shi, Z.J., Zhao, P., Yang, J., 2022. Effect of different washing methods on reducing the inhibition of surface lignin in the tetraethylammonium chloride/oxalic acid-based deep eutectic solvent pretreatment. Ind. Crop. Prod. 188, 115728 https://doi.org/10.1016/j.indcrop.2022.115728.

Yu, H., Xue, Z., Lan, X., Liu, Q., Shi, R., Mu, T., 2020. Highly efficient dissolution of xylan in ionic liquid-based deep eutectic solvents. Cellulose 27, 6175–6188. https://doi.org/10.1007/s10570-020-03213-7.

Zhang, Z.P., Lu, J., Pan, Q.W., Cheng, Y., Tao, Y.H., Du, J., Wang, H.S., 2023. Highly efficient pretreatment at low enzyme loading from reed: Synergy of liquid hot water–ammonia/oxygen and non-ionic surfactants as enzymatic hydrolysis additive. Ind. Crop. Prod. 193, 116257 https://doi.org/10.1016/j.indcrop.2023.116257.

Zheng, W., Lan, T., Li, H., Yue, G., Zhou, H., 2020. Exploring why sodium lignosulfonate influenced enzymatic hydrolysis efficiency of cellulose from the perspective of substrate-enzyme adsorption. Biotechnol. Biofuels 13, 19. https://doi.org/10.1186/s13068-020-1659-5.

Zhou, J.M., Wu, S.B., Liu, Y., 2024. Structural changes of poplar lignin during the ternary deep eutectic solvent (DES) treatment and synergetic alkali-DES treatment. Ind. Crop. Prod. 208, 117782 https://doi.org/10.1016/j.indcrop.2023.117782.